

Measuring AI Ability to Complete Long Software Tasks

50% Task-Completion Time Horizon Doubling Every ~7 Months

Thomas Kwa, Ben West, Joel Becker et al. — METR (Model Evaluation & Threat Research)

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A New Metric: 50% Task-Completion Time Horizon

1

Assemble Task Suite

170 diverse software &
research tasks (3 sec – 8 hrs)



2

Measure Performance

Time skilled humans;
measure AI success rates



3

Fit Logistic Model

Estimate task duration at
which AI succeeds 50%

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

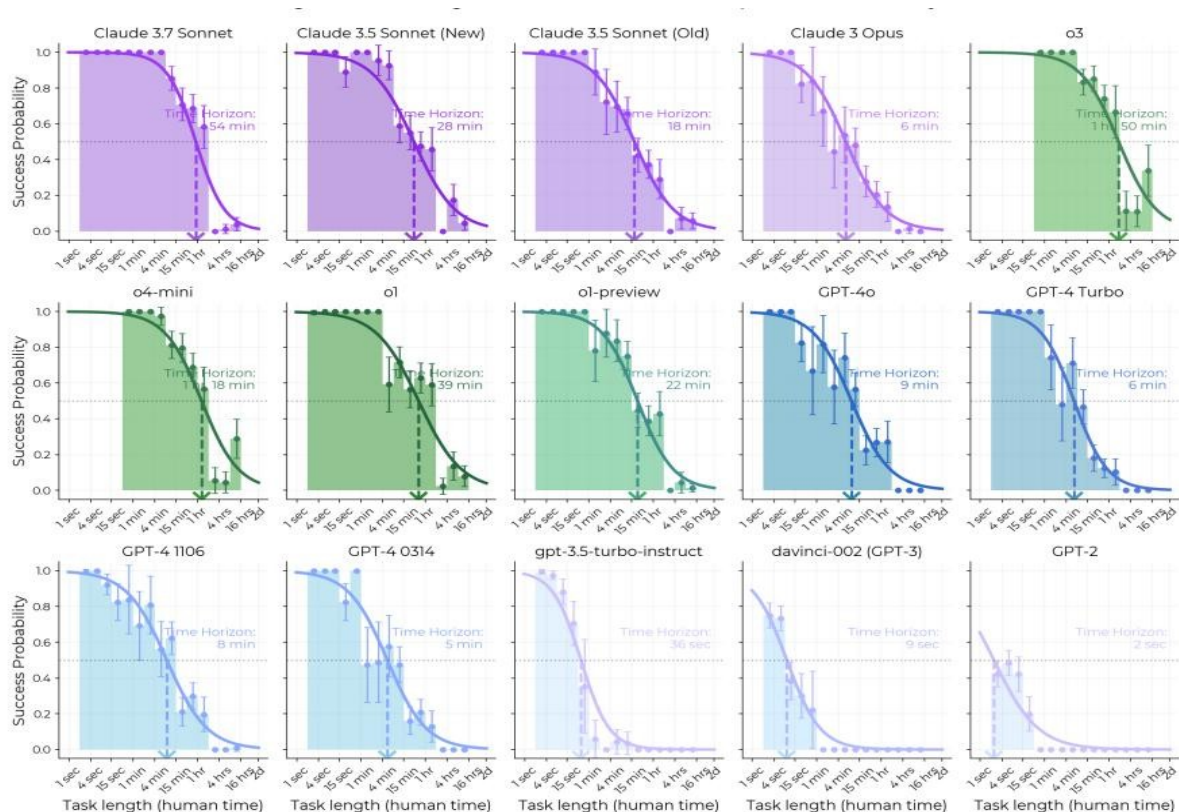
Suite	Tasks	Human Time Range	Description
HCAST	97	1 min – 30 hrs	Diverse software engineering tasks
RE-Bench	7	8 hrs (all)	Difficult ML research engineering tasks
SWAA	66	1 sec – 30 sec	Short single-step atomic actions

Example Tasks

Task	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation
munge_data	56 minutes	Write Python script to transform JSON data
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup

800+ human baselines • 2,529 total hours collected

AI Task Horizon: 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years



2023-2025 growth rate ~20% faster than overall trend

~207 days

doubling time (~7 months)

$R^2 = 0.97$

over 6 years of model releases

GPT-2 (2019)

~2 seconds

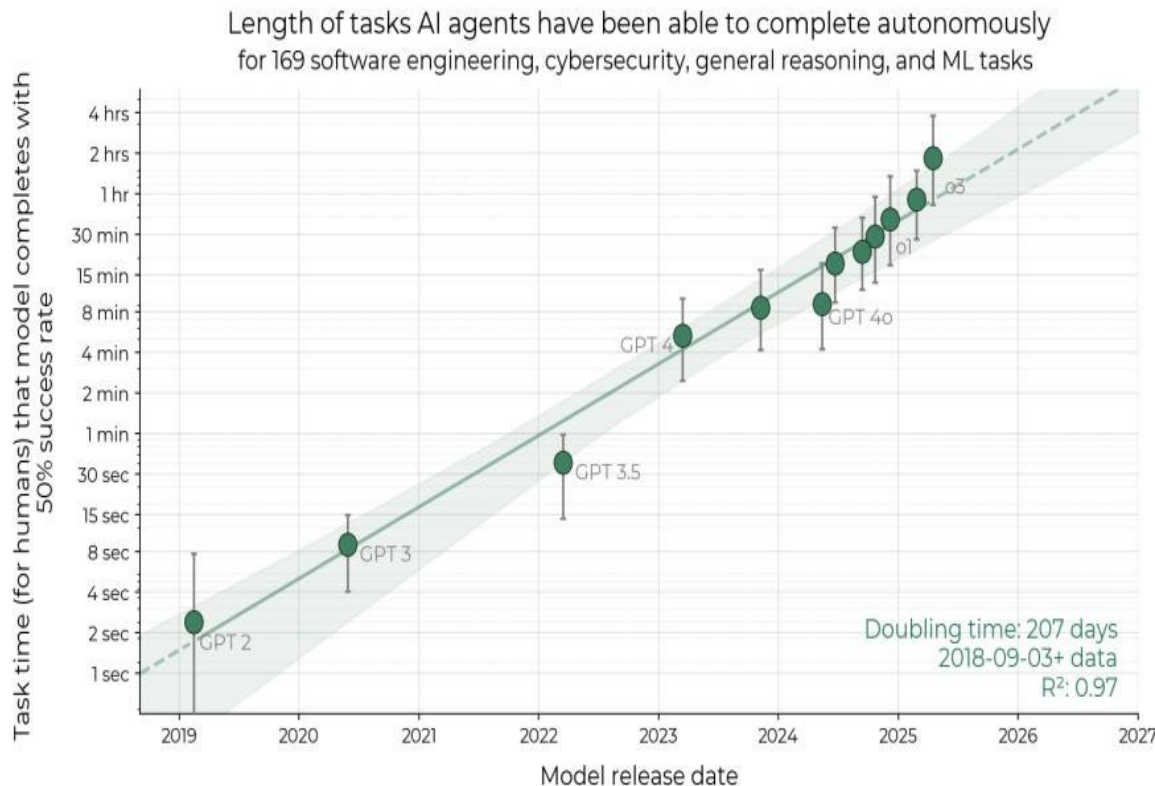
GPT-4 (2023)

~5 minutes

o3 (2025)

~1 hr 50 min

Success Probability Drops Sharply as Task Length Increases



Source: Kwa et al. (2025), Figure 3

Key Observations

- o3**
~1 hr 50 min — Highest 50% horizon
- Claude 3.7 Sonnet**
~54 minutes — Strong frontier performer
- o1**
~39 minutes — First reasoning model
- GPT-4**
~5 minutes — 2023 breakthrough
- GPT-2**
~2 seconds — 2019 baseline

Frontier models push the success curve rightward, solving longer tasks at higher rates.

Reasoning, Tool Use, and Error Recovery Drive Growth

Improved Logical Reasoning

Models demonstrate increasingly sophisticated multi-step reasoning chains, enabling them to decompose complex problems.

Better Tool Use

Effective use of code execution, file manipulation, and external tools to complete software tasks autonomously.

Reliability & Self-Awareness

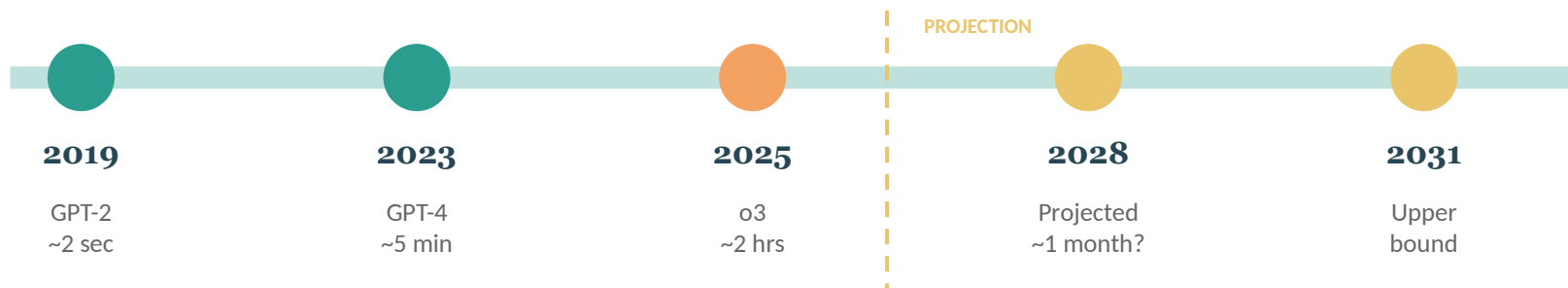
Greater consistency in task execution with improved ability to recognize limitations and uncertainty.

Error Recovery & Adaptation

Ability to detect mistakes, adjust strategies mid-task, and recover from failures without human intervention.

Limitation: Performance is notably lower on less structured, "messier" tasks. External validity to real-world intellectual labor remains an open question.

Extrapolation: Month-Long Task Automation by 2028–2031



167

work hours (1 month)

Projected 50% time horizon if the exponential trend continues at current pace. Reached between mid-2028 and mid-2031 depending on rate assumptions.

~20%

faster recent growth

2023–2025 growth rate is approximately 20% faster than the overall 2019–2025 rate, suggesting possible acceleration of the trend.

Caveats: External validity uncertain; tasks may not represent average real-world intellectual labor. Trend could decelerate.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

- 1 50% time horizon is a robust, intuitive metric for tracking AI autonomy growth — $R^2 = 0.97$ over 6 years of model releases.
- 2 Doubling every ~7 months with possible acceleration since 2024 — the 2023–2025 rate is ~20% faster than the overall trend.
- 3 Current frontier at ~2 hours of human-equivalent autonomous work (o3), up from 2 seconds in 2019.
- 4 External validity remains an open question — benchmark tasks may not perfectly represent real-world intellectual labor.
- 5 Key limitations include poor performance on unstructured tasks and uncertain generalization to real-world job distributions.